

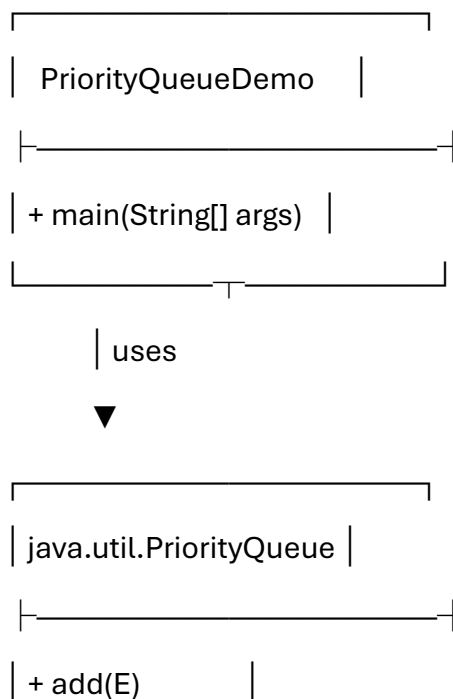
Aim

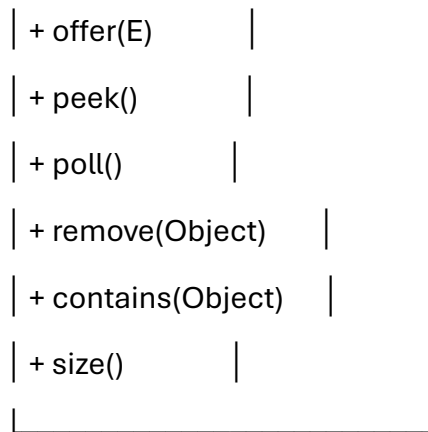
To write a Java program to demonstrate the important methods of the **PriorityQueue** class, such as add(), offer(), peek(), poll(), remove(), contains(), and size().

Algorithm

1. Start the program.
2. Import the java.util.PriorityQueue class.
3. Create a PriorityQueue object.
4. Insert elements using add() and offer().
5. Display the priority queue.
6. Use peek() to retrieve the head element without removing it.
7. Use poll() to retrieve and remove the head element.
8. Use remove() to remove a specified element.
9. Use contains() to check whether an element exists.
10. Use size() to find the number of elements.
11. Display all results.
12. Stop the program.

UML Diagram





Code:

```
package Observations;
```

```
import java.util.PriorityQueue;
```

```
public class PriorityQueueClass {
```

```
    public static void main(String[] args) {
```

```
        // Creating PriorityQueue
```

```
        PriorityQueue<Integer> pq = new PriorityQueue<>();
```

```
        // add()
```

```
        pq.add(30);
```

```
        pq.add(10);
```

```
        pq.add(50);
```

```
        // offer()
```

```
        pq.offer(20);
```

```
        pq.offer(40);
```

```
        System.out.println("PriorityQueue: " + pq);
```

```
// peek()

System.out.println("Head using peek(): " + pq.peek());


// contains()

System.out.println("Contains 30: " + pq.contains(30));


// remove(Object)

pq.remove(30);

System.out.println("After remove(30): " + pq);


// poll()

System.out.println("Removed head using poll(): " + pq.poll());

System.out.println("After poll(): " + pq);


// size()

System.out.println("Size of PriorityQueue: " + pq.size());

}

}
```

Output:

```
es' '-cp' 'C:\Users\Sahithi_pyla\AppData\Roaming\Cod
bservations.PriorityQueueClass'
PriorityQueue: [10, 20, 50, 30, 40]
Head using peek(): 10
Contains 30: true
After remove(30): [10, 20, 50, 40]
Removed head using poll(): 10
After poll(): [20, 40, 50]
Size of PriorityQueue: 3
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>
```

Result:

Thus, the Java program successfully demonstrates the PriorityQueue methods add(), offer(), peek(), poll(), remove(), contains(), and size().